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OCNR: Stabilizing Self-Play by Mitigating Iteration-Collapse with One-Class Novelty Rewards

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Background

Post-Training via Closed-Loop Self-Play

Reinforcement Learning improves LLM reasoning capabilities, in practice.

Data Bottleneck: Human-curated data is expensive, finite, and hard to scale.

SELF-PLAY CYCLE



Questioner

Creates novel questions



Solver

Generates answers

Goal: open-ended self-improving loop achieving (near) zero-data continuous scaling.

Background

The Wall: Iteration Collapse

Self-Play pipelines offer a path toward open-ended improvement, in principle.

Iteration Collapse: Performance initially improves but subsequently degrades.

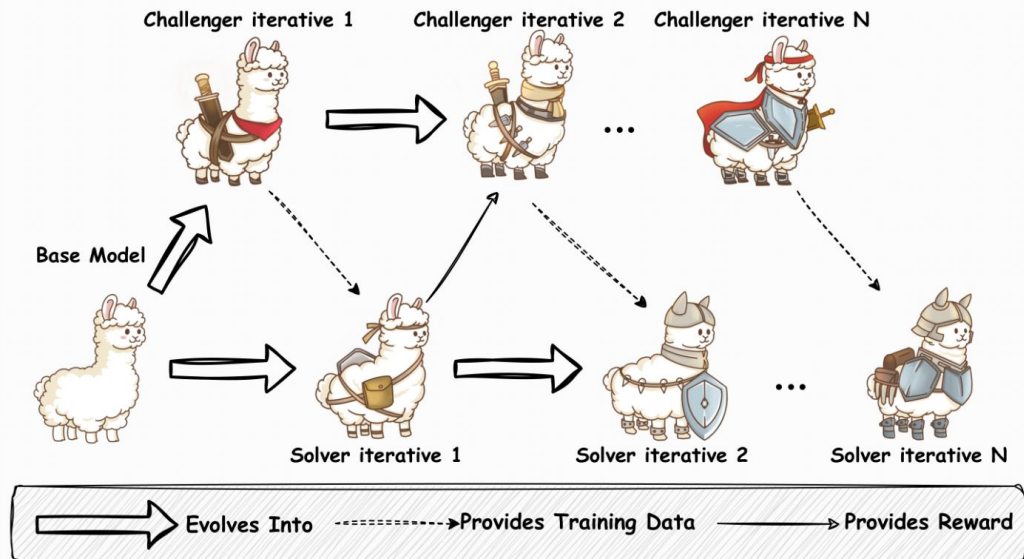


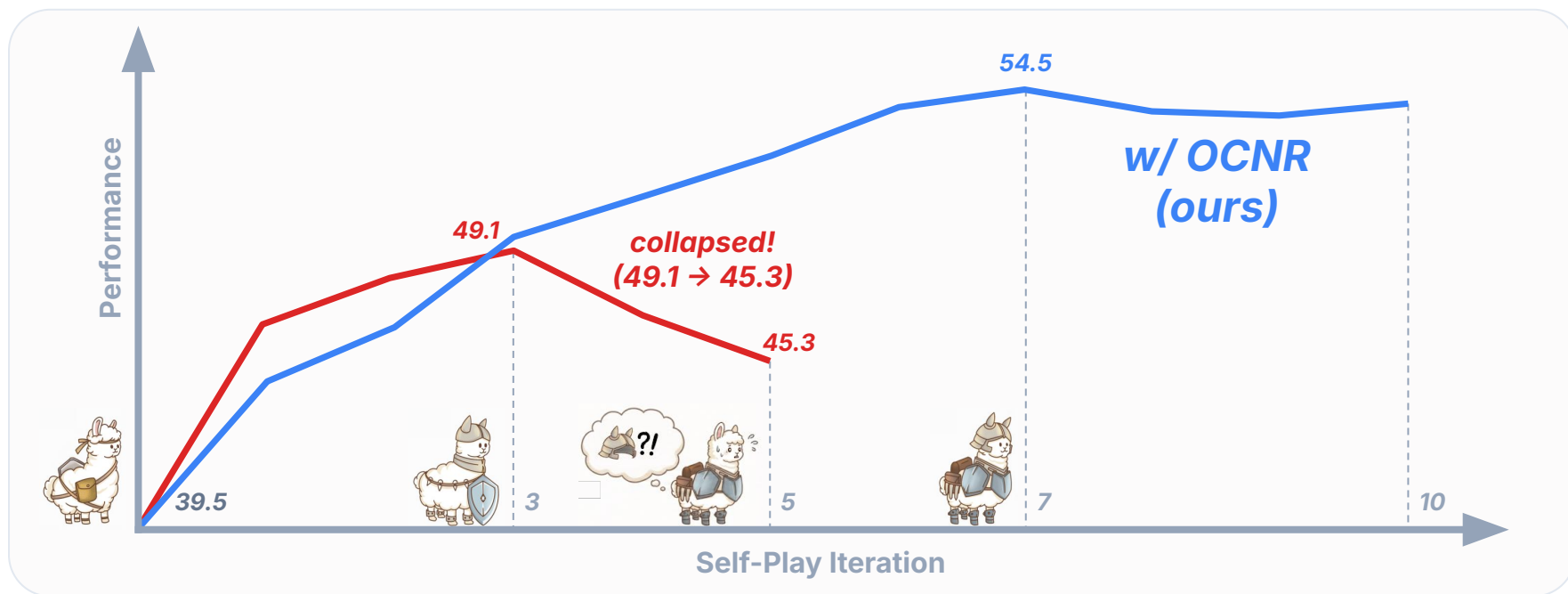
Image credit: R-Zero (Huang et al., 2026)

Background

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Self-Play pipelines offer a path toward open-ended improvement, in principle.

Iteration Collapse: Performance initially improves but subsequently degrades.



Approach

Novelty as the Key to Stability

(1) Validity

Questions should be solvable and well-formatted.

(2) Difficulty

Questions should be tailored to the solver's level.

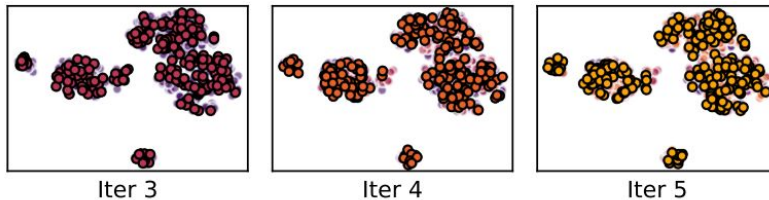
CRITICAL

(3) Diversity

Questions should encompass a wide range of types.

Previous attempts for the diversity

- Explicit prompting (Absolute Zero; *Zhao et al., 2025*).
- Batch-level BLEU penalties (R-Zero; *Huang et al., 2026*).



2D t-SNE of question embeddings across self-play iteration

Problems are diverse within an iteration, but non-diverse across iterations.

The core issue is not just diversity but novelty: problems must be genuinely new from the solver's perspective.

Approach

Novelty Quantification

Seen Detector (via Learnable Prototype Vector)

$$\text{SEEN}(q_t) = \exp \left(-\frac{\|h(q_t) - w_s\|^2}{\tau} \right)$$

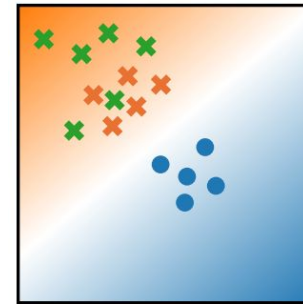
One-Class Novelty Reward (OCNR)

$$R_Q^{\text{nov}}(q_t) = 1 - \text{SEEN}(q_t)$$

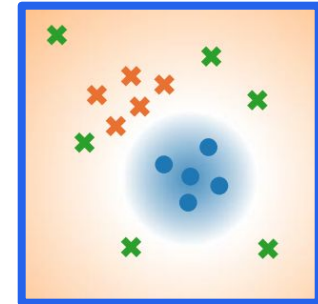
Conceptual illustration of novelty rewards

Avoids global half-space penalties of sigmoid boundaries!

● Seen × Unseen × To be generated

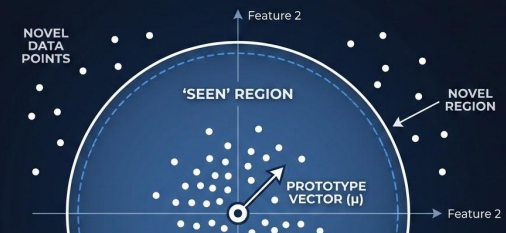


(a) Sigmoid-based



(b) Distance-based

*OCNR penalizes redundant tasks while targeting the solver's **actual blind spots**.*



Approach

The Alternating Self-Play Loop

01

Generate & Split Questions

Construct a balanced **Seen/Unseen** sets.

02

Update Solver (baseline reward)

Train the solver using only the **Seen** subset.

03

Update Detector (binary cross-entropy)

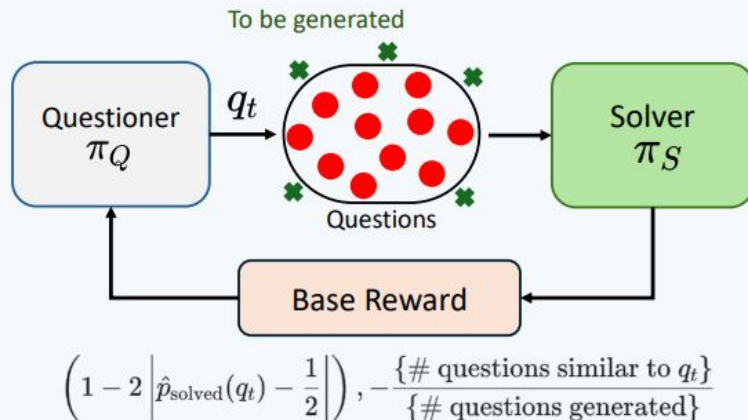
Train the detector on **Seen/Unseen** sets.

04

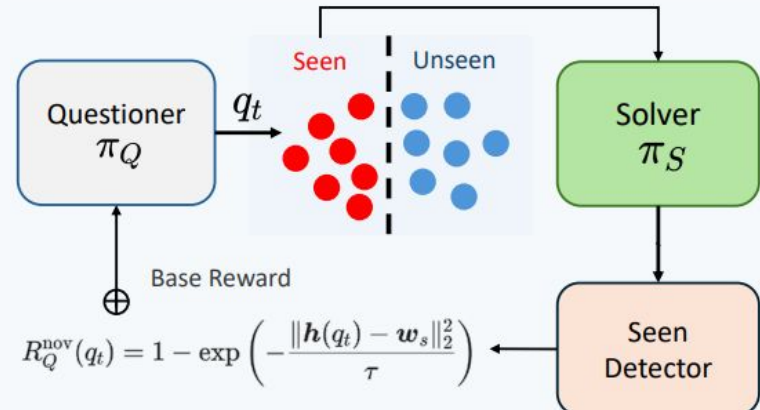
Update Questioner (baseline + novelty reward)

Update the Questioner by adding the OCNR novelty term to its base reward.

Original Pipeline (R-Zero)



OCNR Pipeline (R-Zero w/ OCNR)



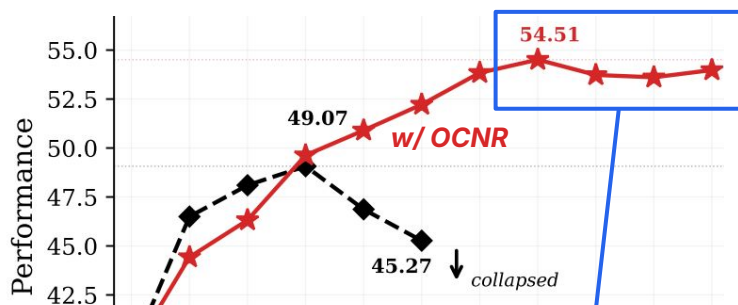
Main Results

Mathematical Reasoning

Method	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5
R-Zero	46.5	48.1	49.1	46.9	45.3
w/ OCNR (Ours)	44.4	46.3	49.6	50.9	52.2

10-Iteration Extension

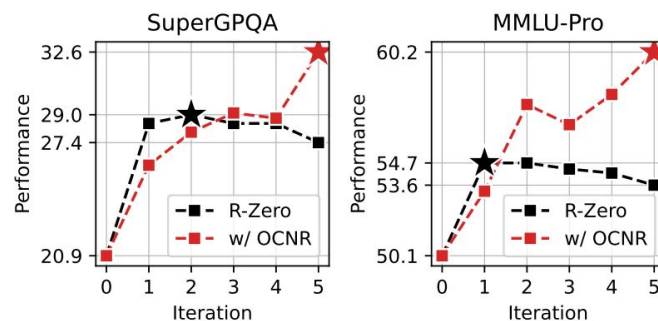
Extended testing demonstrates that OCNR sustains improvement up to **Iteration 7**.



Notably, it maintains **the stable plateau** through Iteration 10, preventing post-peak degradation.

Cross-Domain Leap

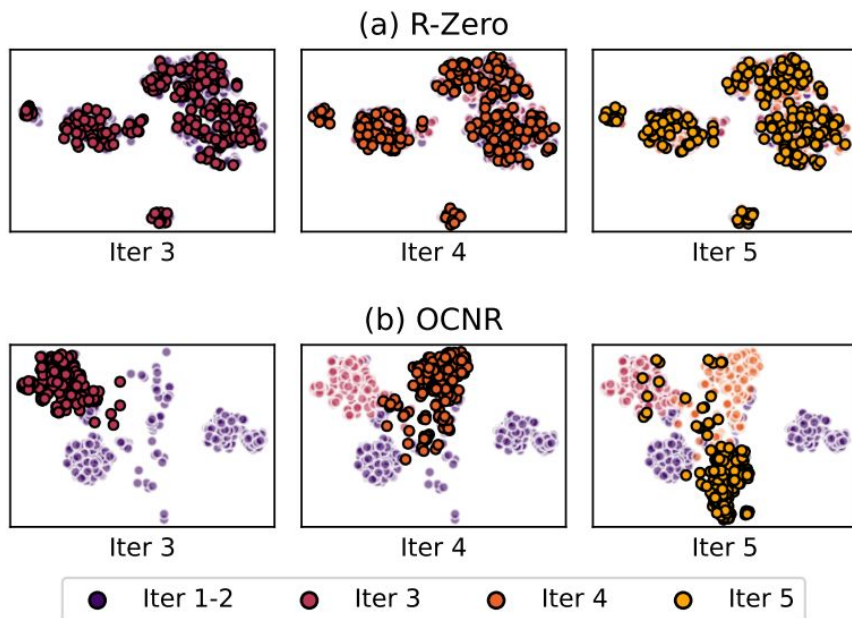
Math-centric self-play successfully transfers to **general-purpose reasoning** datasets at test time.



Notably, it enhances the self-play pipeline itself, moving **beyond task-specific gains**.

Analytical Results

Diversity and Novelty of Questions



2D t-SNE of question embeddings across self-play iteration

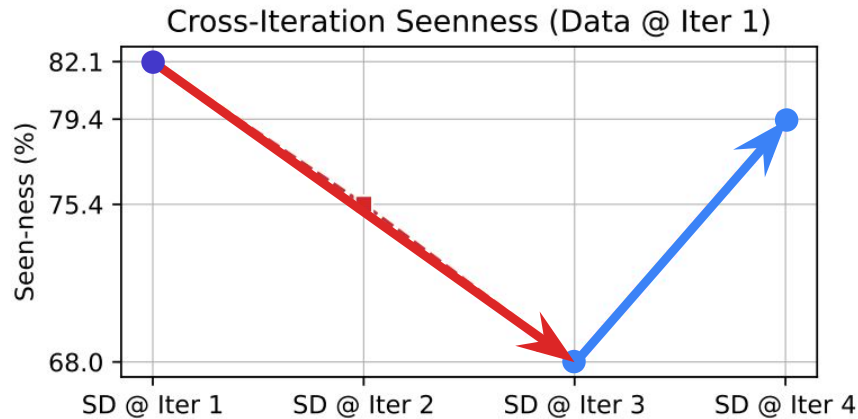
Non-diverse across iterations!

Diverse across iterations!

Our strategy yields not only diverse but specifically novel questions.

Analytical Results

Re-emergence of Forgotten Questions



Question

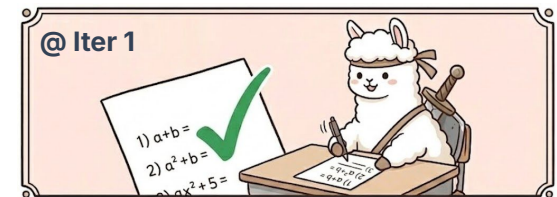
Let a, b, c be the roots of the cubic polynomial $x^3 - 5x^2 + 6x - 1 = 0$.

Find the value of the expression $(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$.

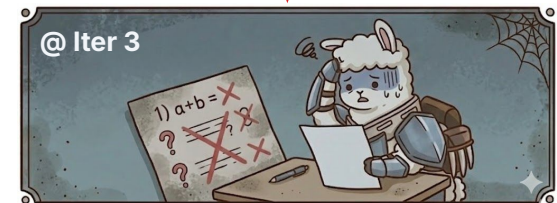
Re-emerged

Let x_1, x_2, x_3 , and x_4 denote the roots of the equation: $x^4 - 6x^3 + 5x^2 - 4x + 2 = 0$

Determine the value of the sum: $\sum_{k=1}^4 \frac{1}{x_k}$



forgetting questions
as time goes by...



brushing up on
forgotten questions!

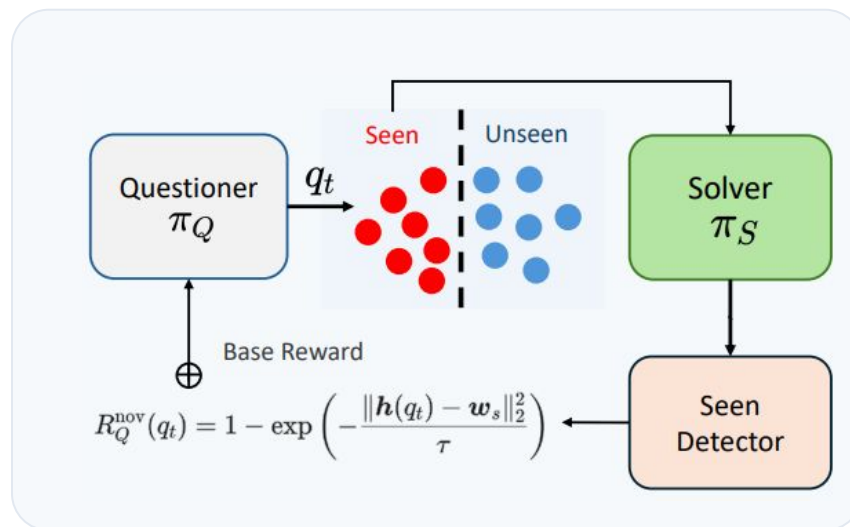
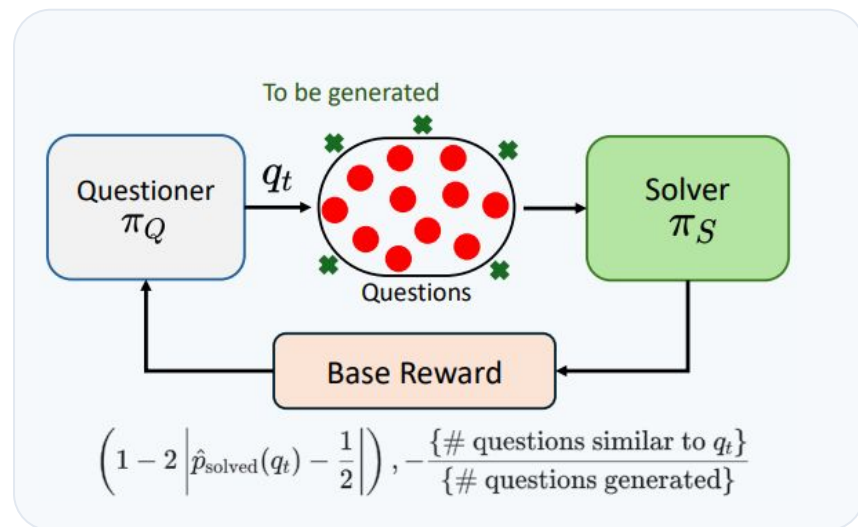


Our strategy makes a questioner re-discovers forgotten concepts.

Analytical Results

Over-generation of Questions

Method	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5
R-Zero	46.5	48.1	49.1	46.9	45.3
R-Zero (2x)	47.5	49.2	47.8	46.1	44.0
w/ OCNR (Ours)	44.4	46.3	49.6	50.9	52.2



Our strategy goes beyond the simple benefits of over-generation.

Summary and Open Directions

- **Core Finding:** iteration collapse is fundamentally caused by cross-iteration distribution degeneration into familiar spaces.
- **Proven Solution:** OCNr stops redundancy at the source, offering a simple, computational-efficient plug-in for data-free scaling.
- **Future Extensions:** moving beyond our memory-based single prototype approach; improving sample efficiency; integrating safety guardrails.

Thank you!